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30 APRIL 1957
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9 JANUARY 1950

MILITARY SPECIFICATION

INDICATOR, TURN AND SLIP, 28V., DC, 2-3/4 SIZE, TYPE MD-10

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers certain design requirements and all performance requirements for procurement of 2¾ inch, 28 volt, direct current, turn and slip indicators.

1.2 Classification. This specification covers a standard 2 minute turn and slip indicator, Type MD-10, MS Part No. 25069.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-C-320 — Chromium Plating (Electrodeposited).

QQ-N-290 — Nickel Plating (Electrodeposited).

QQ-P-416 — Plating Cadmium (Electrodeposited).

MILITARY

MIL-A-8625 — Anodic Coatings, for Aluminum and Aluminum Alloys.

MIL-C-5015 — Connectors; Electrical, "AN" Type.

MIL-D-5028 — Drawings and Data Lists; Preparation of Manufacturer's (For Production Aircraft, Guided Missiles, Engines, Accessories and Other Auxiliary Equipment).

MIL-E-5272 — Environmental Testing, Aeronautical and Associated Equipment, General Specification for.

MIL-I-6181 — Interference Limits and Tests, Aircraft Electrical and Electronic Equipment.

MIL-N-6880 — Lubrication of Aircraft; General Specification for.

FSC 6610

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MIL–N–3356 — Nut; Self Locking, Instrument Mounting.

MIL–P–6064 — Packaging of Lightweight Aircraft Accessories.

MIL–P–6906 — Plates; Information and Identification.

MIL–S–7742 — Screw - Threads; Standard, Aeronautical.

STANDARDS

MILITARY

MIL–STD–129 — Marking for Shipment and Storage.

MIL–STD–130 — Identification Marking of U. S. Military Property.

MS–28069 — Indicator, Turn and Slip, 28 Volt Direct Current.

DRAWINGS

AIR FORCE — NAVY AERONAUTICAL STANDARD DRAWINGS

AN5727 — Glass, Aircraft Instrument Cover.

AND10398 — Metals—Definition of *Dissimilar*.

AND10400 — Numerals and Letters —Aircraft Instrument Dial, Standard Form of.

PUBLICATIONS

AIR FORCE — NAVY AERONAUTICAL SPECIFICATION BULLETIN

No. 143 — Specifications and Standards, Use of.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The indicator furnished under this specification shall be a product which has been tested and has passed the Qualification tests specified herein.

3.2 Materials. Materials shall conform to applicable specifications and as specified herein. Materials which are not covered by applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.2.1 *Critical Material.* Non-critical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.2.2 *Metals.* Metals shall be of the corrosion-resistant type, or suitably protected to resist corrosion in salt spray or atmospheric conditions to which the indicator may be subjected when in storage or during normal service life.

3.2.2.1 *Dissimilar Metals.* Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in Drawing AND10398.

3.2.3 *Nonmagnetic Materials.* Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

3.2.4 *Fungus-Proof Materials.* Materials which are not nutrients for fungi shall be used to the greatest extent practicable. Where materials which are nutrients for fungi must be used, such materials shall be treated with

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a fungicidal agent, as approved by the procuring activity.

3.2.5 *Selection of Materials.* Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.2.5.1 *Standard Parts.* Standard parts (AN or MS) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the (AN or MS) standard parts without alteration, and provided the corresponding standard part numbers are referenced in the parts lists and, if practicable, on the contractor's drawings. In case there is no suitable corresponding AN or MS standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 **Design and Construction.** The indicator shall be designed to indicate the rate of turn of an airplane about its vertical axis and the lateral attitude of an airplane relative to the apparent vertical. The design shall incorporate means to prevent fogging on the surface of the cover glass under extremes of atmospheric conditions. The means provided shall be subject to approval by the procuring activity. The design shall be reasonably simple, to permit overhaul or repair without requiring special tools and fixtures. The indicator shall be constructed to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation and service.

3.3.1 *Case.* The indicator case shall conform to Standard Drawing MS 28069. The case shall suitably protect the mechanism against the infiltration of dust and sand particles.

3.3.1.1 *Body.* The body shall be made either of aluminum alloy or of synthetic material, be uniform in texture, and have a smooth surface with a durable dull black finish. Synthetic material shall be of the thermosetting type composed of a suitable filler and phenol condensation product binder.

3.3.1.2 *Bezel Ring.* The bezel ring shall be made of aluminum alloy and shall have a durable dull black finish. The bezel ring shall be held in place by means of screws. Provision shall be made for replacement of the cover glass by removal of the bezel ring.

3.3.1.3 *Cover Glass.* The cover glass shall be clear and free from flaws which interfere with normal reading of the indicator. The diameter of the glass shall conform to AN5727.

3.3.2 *Mounting.* The indicator shall be suitable for front or rear mounting on the instrument panel.

3.3.3 *Dial.* The dial shall consist of a turn indicator and a slip indicator.

3.3.3.1 *Turn Indicator.*

3.3.3.1.1 *Pointer.* The turn indicator pointer and center mark shall conform to Figure 1. The pointer shall be firmly attached to the mechanism, shall be as light as practicable, and shall be sufficiently rigid to prevent oscillation under vibration. The pointer shall pass through the dial face slot which shall be designed to conceal the internal mechanism. Means shall be provided for readily adjusting the pointer to the zero position.

3.3.3.1.2 *Mechanism Adjustment.* Means shall be provided to adjust or correct the sensitivity and the damping of the indicator. When the adjustment means is set in the neutral position, one quarter part of the total range of adjustment shall be available in

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either direction. This means of adjustment shall be reasonably simple to manipulate without special tools.

3.3.3.2 *Slip Indicator.*

3.3.3.2.1 *Tube.* The slip indicator tube shall be clear annealed glass tubing having a wall thickness of not less than 3/64 inch and shall be free from any flaws that will seriously affect the readability. The inside of the tube shall be smooth and uniform so that the indicator ball will roll smoothly.

3.3.3.2.2 *Ball.* The ball shall be of white glass or other material approved by the procuring activity. It shall be a highly polished accurate sphere $\frac{3}{8} \pm \frac{1}{64}$ of an inch in diameter. The white color shall be in accordance with the following Munsell System color notation; neutral, not less than 9.0 value and no greater than 0.5 Chroma.

3.3.3.2.3 *Damping Liquid.* The damping liquid used in the slip indicator shall be sufficiently colorless so as not to interfere with satisfactory silhouette illumination of the ball by the dull black backing of the indicator tube.

3.3.4 *Electrical Receptacle.* An electrical receptacle shall be provided at the rear of the case and shall be of a design in accordance with Specification MIL–C–5015, and as shown on standard Drawing MS 28069.

3.3.5 *Wiring.* The electrical wiring system shall be insulated from the indicator case. Pins A and B of the receptacle shall be utilized as the positive and negative leads, respectively, of the 28 volt DC power.

3.3.6 *Filters.* Filters used to reduce radio interference shall be acceptable to the procuring agency.

3.4 **Performance.** The turn and slip indicator shall satisfy the performance requirements when subjected to the applicable tests as specified in section 4.

3.5 **Interchangeability.** All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL–D–5028.

3.6 **Special Tools.** The design of the turn and slip indicator shall be such as to accommodate to the greatest possible extent, disassembly, reassembly, and service maintenance by those tools and items of maintenance equipment which are normally available as commercial standards.

3.7 **Physical Requirements.**

3.7.1 *Weight.* The weight of the indicator shall not exceed 2 pounds.

3.7.2 *Finish.* Protective coatings and finishes which will crack, chip or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

3.7.2.1 *Aluminum Alloy Parts.* Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL–A–8625.

3.7.2.2 *Iron and Steel Parts.* Where practicable, iron and steel parts shall be chromium, cadmium or nickel plated in accordance with Specifications QQ–C–320, QQ–P–416 and QQ–N–290, respectively.

3.7.3 *Lubrication.* Lubrication shall be accomplished in accordance with Specification MIL–I–6880, except that the selection of the lubricants used shall be approved by the Bureau of Aeronautics.

3.7.4 *Screw Threads.* Screw threads 0.060 inch or larger in diameter shall be in accordance with Specification MIL–S–7742.

3.8 **Marking.** All markings shall be durable to withstand usage encountered in service.

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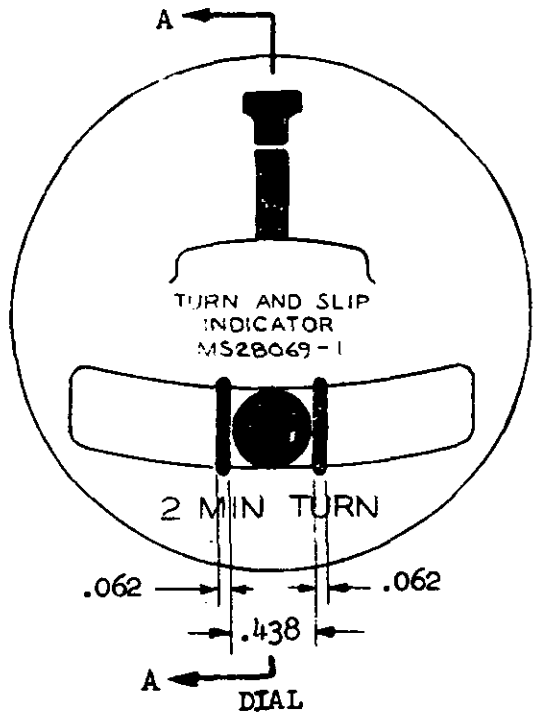
The form of the letters and numerals shall conform to Drawing AND10400.

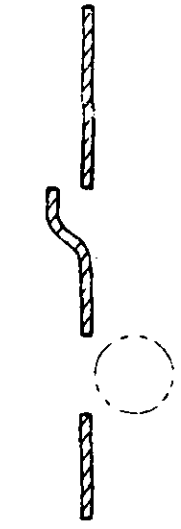
3.8.1 *Dial.* The dial shall be marked as shown in Figure 1.

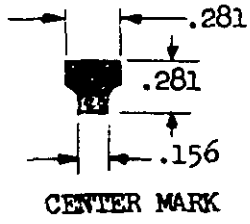
3.8.1.1 *Visibility of Dial.* The pointer, centermark and other specified markings on the dial shall be visible from any point within the frustrum of a cone whose side makes an angle of 30 degrees with a perpendicular to the dial and whose small diameter is the aperture of the instrument case.

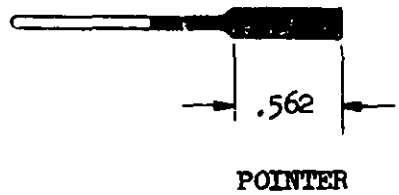
3.8.2 *Identification of Product.* The indicator shall be marked for identification in accordance with MIL–STD–130. The nameplate shall contain the following information:

Indicator, Turn and Slip, 28V DC, 2¾ Size, Type MD–10
MS Part No. 28069–1
Stock No.
Manufacturer's Part No.
Manufacturer's Name or Trademark
U. S.









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Dimensions in inches, unless otherwise specified, tolerances: decimals ± 0.16

Marking	Height or length	Width of line or graduation ± .005	Material or Finish
Center Mark Shaded Portion of Pointer Lines Marking Zero Position of Ball Lettering "2 Min Turn"	As shown on figure .156	As shown on figure .031	Durable Matte White
Rear Half and Bottom of Slip Indicator or Suitable Card Behind Inclinometer	----	----	Durable
Unshaded Portion of Pointer	....	....	Dull
Background of Dial	....	....	Black
Lettering			
"Turn and Slip"	.047	.016	
"Indicator"	.047	.016	
"MS 28069–1"	.047	.016	

FIGURE 1. *Dial markings.*

3.8.2.1 *Use of MS or MIL Designations.* MS or MIL designations shall not be applied to a product except for Qualification test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification, or from the Aeronautical Standards Group.

3.8.3 *Power Identification.* The following markings in durable dull white or natural metal finish shall be permanently and legibly marked on the back of the indicator case beside the electrical receptacles, to identify the electrical connector pin with which they shall be radially aligned:

\+ Aligned with Pin "A"

— Aligned with Pin "B"

3.9 Installation Instruction. The contractor shall furnish with each indicator one printed copy of instructions, with illustrations and diagrams, if necessary, covering the installation of the indicator. Prior to printing, two copies shall be furnished to the Procuring Activity for approval. The instructions shall be printed on 8½ by 11 or 11 by 17 inch paper.

3.9.1 *Envelope.* An envelope furnished by the contractor containing installation instructions, mounting nuts, and mounting screws, (see Drawing MS 28069) shall be packaged with each indicator, and shall be marked with the following information:

"IMPORTANT"

"THIS ENVELOPE CONTAINS INSTRUCTIONS, MOUNTING NUTS, AND MOUNTING SCREWS"

3.10 Workmanship. All details of workmanship shall be in accordance with high grade aircraft instrument manufacturing practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests. The inspection and testing of the indicators shall be classified as follows:

(a) Qualification Tests: Qualification tests are those tests performed on indicators submitted for qualification as a satisfactory product.

(b) Acceptance Tests: Acceptance tests

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are those tests performed on indicators manufactured and submitted for acceptance under contract.

4.2 **Qualification Tests.** The qualification tests of the indicators shall consist of all tests of this specification and examination of product. Qualification tests may, at the option of the qualifying agency, be supplemented with tests under actual service conditions. The tests shall be conducted in the following sequence:

Individual Tests
Representative Tests
Radio Interference
Low Temperature-High Altitude Exposure
High Temperature Exposure
Vibration Failure
Humidity
Fungus Resistance
Salt Spray
Mounting Lugs
Life

4.2.1 *Qualification Test Samples.* Qualification test samples shall consist of two turn and slip indicators of each manufacturer's part number upon which qualification is desired. Samples shall be identified as specified herein and forwarded to the activity responsible for qualification, designated in the letter of authorization from activity.

Sample for Qualification Test
Submitted by (Name) (Date) for Qualification Tests in accordance with the requirements of specification MIL–I–5488A under authorization (reference letter authorizing test)
Indicator; Turn and Slip, 28V., DC, Type MD–10
Manufacturer's Part No.
Name of Manufacturer

4.3 **Acceptance Tests.** The acceptance tests of the indicators shall consist of the individual tests and the representative tests of this specification.

4.3.1 *Individual Tests.* The individual tests of the indicators shall consist of the following tests conducted on each indicator:

Examination of Product
Low Voltage Operation
Current Consumption
Turn Indicator Static Balance
Turn Indicator Sensitivity
Turn Indicator Damping
Turn Indicator Dynamic Balance
Slip Indicator Visibility
Slip Indicator Filling
Slip Indicator Zero Position
Slip Indicator Friction
Slip Indicator Damping
Slip Indicator Sensitivity

4.3.2 *Representative Tests.* The representative tests of the indicators shall consist of the following tests conducted on each representative test sample:

Magnetic Effect
Dielectric Strength
Vibration Error
Low Temperature Operation
  Turn Indicator Sensibility (Low Temperature)
  Slip Indicator Damping (Low Temperature)
High Temperature Operation
  Turn Indicator Sensitivity (High Temperature)

4.3.2.1 *Representative Test Samples.* Representative test samples shall consist of at least one indicator selected at random by the Inspector from each lot of 100 or less indicators which have passed the individual tests. A lot shall consist of identical indicators with the same manufacturer's part number, manufactured under substantially the same conditions and submitted at the same time. The Inspector shall select as many indicators as considered necessary to ensure compliance of the entire lot with the requirements of these tests. Indicators passing the representative tests shall not be delivered on contract until they have been rebuilt and resubmitted to all individual tests.

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4.4 Test Conditions.

4.4.1 *Atmospheric Conditions.* Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of approximately 29.92 inches of mercury and at a temperature of approximately 25°C (77°F). When tests are made with either, or both, the atmospheric pressure and temperature substantially different from the above values, proper allowance shall be made for the difference from specified conditions.

4.4.2 *Vibration Stand.* Whenever a vibration stand is specified, it shall consist of a device which will vibrate at a desired frequency between 300 and 3,000 cycles per minute and shall subject the indicator to such vibration that a point of the vibration stand shall describe in a plane inclined 45 degrees to the horizontal plane, a circle of the diameter specified herein.

4.4.3 *Level Position.* Whenever level position is specified, the instrument shall be in a position in which the upper mounting holes are on a horizontal line and the face is vertical.

4.4.4 *Electrical Power.* Unless otherwise specified, tests requiring the operation of the indicator shall be conducted utilizing a 28 volt, direct current power supply.

4.4.5 *Tapping.* Unless otherwise specified, the indicator may be gently tapped or vibrated before a test reading is taken.

4.5 Test Methods.

4.5.1 *Examination of Product.* Each indicator shall be examined externally to determine conformance with all the requirements of this specification not covered by tests.

4.5.2 *Low Voltage Operation.* The electrical power, with the voltage reduced to a value not exceeding 22 volts, shall be applied to the indicator. The indicator shall start and continue to operate.

4.5.3 *Current Consumption.* A calibrated direct current ammeter and voltmeter shall be connected in the line between the indicator and the source of electric power. After the indicator has operated for a period of 3 minutes, the current consumption shall not exceed 0.15 ampere with an electrical potential of 28 volts at the indicator terminals.

4.5.4 *Turn Indicator Static Balance.* The indicator, with no power applied, shall be placed successively in the level, 90 degree bank and 90 degree dive positions, and the pointer position relative to the center mark shall be checked. The deviation of the pointer from the zero turn position shall not exceed $\frac{1}{32}$ of an inch.

4.5.5 *Turn Indicator Dynamic Balance.* The indicator shall be operated for 3 minutes and then be placed successively in the level, 90 degree bank, and 90 degree dive positions, and the pointer position relative to the center mark shall be checked. The deviation of the pointer from the zero turn position shall not exceed $\frac{1}{32}$ of an inch.

4.5.6 *Turn Indicator Sensitivity.* The indicator shall be mounted in the level position on a rate table and operated for 3 minutes. The indicator shall then be rotated about a vertical axis at the rates of turn specified in Table 1. The movement of the pointer shall be smooth throughout the range of operation and within the tolerance specified in Table 1. The test shall be performed in both direction of rotation.

TABLE I. *Turn indicator sensitivity*

Rate of turn (Degrees/minute)	Deflection of pointer tip (Arc in inches)
90	$\frac{5}{64} \pm \frac{1}{32}$
180	$\frac{5}{32} \pm \frac{1}{32}$
360	$\frac{3}{16} \pm \frac{1}{32}$
1080	$\frac{3}{4} \pm \frac{1}{8}$

4.5.7 *Turn Indicator Damping.* The indicator shall be placed in the level position on

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a rate table and operated for 3 minutes. The indicator shall then be rotated about its vertical axis at a rate which causes full-scale deflection of the pointer and then the rotation shall be stopped suddenly. The motion of the pointer shall be so damped that the pointer returns to the zero mark without crossing it in not less than one second, and not more than 3 seconds, and without oscillation or noise other than that encountered in normal operation.

4.5.8 *Slip Indicator Visibility.* The slip indicator shall be viewed from a position 12 inches directly in front of the zero mark with the ball alternately constrained by each end of the tube. In each extreme position not less than one-half of the ball shall be visible.

4.5.9 *Slip Indicator Filling.* The indicator shall be tipped so that all the air in the tube is trapped in the expansion chamber. The indicator shall then be placed in the level position and rotated around its longitudinal axis to a bank angle of 45 degrees and with the expansion chamber end of the tube low. During this rotation, no part of the air bubble shall be visible when viewed from a point 12 inches directly in front of the slip indicator zero mark.

4.5.10 *Slip Indicator Zero Position.* The indicator shall be placed in the level position and the ball shall rest at the zero mark within $\frac{1}{32}$ of an inch.

4.5.11 *Slip Indicator Friction.* From the level position the indicator shall be rotated about its longitudinal axis, alternately to the right and left at a rate of $2 \pm \frac{1}{4}$ degrees per second. The ball shall roll smoothly to the limits of its range and shall not stick at either end of the tube. The indicator shall not be tapped or vibrated during the test.

4.5.12 *Slip Indicator Damping.* From the level position the indicator shall be rotated about its longitudinal axis to a bank angle of 12 degrees. The indicator shall then be suddenly rotated through a vertical angle of 24 degrees, so that the ball will roll to the opposite end of the tube. The time required for the ball to roll from the zero mark to the opposite end of the tube shall not be less than 0.2 seconds.

4.5.13 *Slip Indicator Sensitivity.* From the level position the indicator shall be slowly rotated about its longitudinal axis until the ball reaches the end of the tube or to the limits of its range. At this position the angle of tilt of the indicator shall be measured and shall be within 8 and 12 degrees. This test shall be performed in both directions of rotation.

4.5.14 *Magnetic Effect.* The indicator shall be held in various positions with its nearest part $5\frac{1}{2}$ inches from the center of a short bar-magnet compass whose compensator magnets have been removed, and shall be set up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The various positions shall be so chosen as to include that position which causes the maximum deflection of the compass. The maximum deflection shall not exceed one degree. This test shall be performed first with no voltage applied to the indicator and then shall be repeated with the indicator operating.

4.5.15 *Dielectric Strength.* The completely assembled indicator with filter installed (if filter is used to suppress radio noise), shall be subjected to a potential of 500 volts DC between the positive terminal and conducting parts of the case, and between the negative terminal and conducting parts of the case, each for a period of 5 seconds. There shall be no breakdown of insulation or any other permanent damage to the instrument.

4.5.16 *Vibration Error.* The indicator shall be mounted in a level position on a vibration stand and operated for 3 minutes. The stand shall then subject the indicator to a vibration, as specified herein, with a circle diameter between 0.009 and 0.011 inch, at frequencies between 300 and 3,000 cycles per minute. Throughout this frequency range the maxi-

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mum magnitude of the pointer vibration shall not exceed 1/32 of an inch, and the slip indicator ball shall not differ from its zero position by more than 1/16 of an inch. No natural vibration frequency of the indicator shall be observable at any point in the test frequency range.

4.5.17 *Low Temperature Operation.* (This test may be combined with the low temperature-high altitude exposure test at the discretion of the testing agency). The indicator shall be subjected to an ambient temperature of −55 ± 2°C (−67 ± 3.6°F) for a period of 4 hours. At the end of this period and with the temperature maintained at −55 ± 2°C the indicator shall be subjected to and meet the requirements specified in the turn indicator sensitivity (Low Temperature) and the Slip Indicator Damping (Low Temperature) tests. There shall be no slip indicator leakage as a result of these tests.

4.5.17.1 *Turn Indicator Sensitivity (Low Temperature).* The indicator shall be mounted in the level position on a rate table and operated for 5 minutes. The indicator shall then be rotated about the vertical axis at 360 degrees per minute and the pointer deflection shall be 5/16 inch. The rotation shall then be stopped and the pointer shall return smoothly without oscillation to the zero position within 1/32 of an inch. The test shall be performed in both directions of rotation.

4.5.17.2 *Slip Indicator Damping (Low Temperature).* The Slip Indicator Damping test shall be repeated and the time for the ball to travel from the zero mark to the extreme end of the tube shall be not more than 2½ seconds.

4.5.18 *High Temperature Operation.* (This test may be combined with the High Temperature Exposure test at the discretion of the testing agency). The indicator shall be subjected to an ambient temperature of 70 ± 2°C (158 ± 3.6°F) for a period of 4 hours. At the end of this period, and with the temperature maintained at 70 ± 2°C, the indicator shall be subjected to and meet the requirements specified in the Turn Indicator Sensitivity (High Temperature) test. There shall be no slip indicator leakage as a result of these tests.

4.5.18.1 *Turn Indicator Sensitivity (High Temperature).* The indicator shall be mounted in the level position on a rate table and operated for 3 minutes. The indicator shall then be rotated about the vertical axis at 360 degrees per minute and the pointer deflection shall be 5/16 ± 1/16 inch. The rotation shall then be stopped and the pointer shall return smoothly without oscillation to the zero position within 1/32 of an inch. The test shall be conducted in both directions of rotation.

4.5.19 *Radio Interference.* The electrical system of the indicator shall be tested for conducted and radiated radio noise interference in accordance with Specification MIL–I–6181.

4.5.20 *Low Temperature-High Altitude Exposure.* (This test may be combined with the "Low Temperature Operation" test at the discretion of the testing agency). The indicator, not operating and mounted in a level position, shall be subjected to a temperature of −65 ± 2°C (−85 ± 3.6°F) for a period of 48 hours. During any 4 hours of the 48 hour period, the indicator shall be subjected to a simulated altitude of 50,000 feet, (3.436 inches of mercury absolute pressure). At the end of the 48 hour period, and at atmospheric pressure, the temperature shall be raised to −55 ± 2°C (−67 ± 3.6°F), and the indicator shall be subjected to this temperature for a period of 4 hours. At the end of this period, and with the temperature maintained at −55°C, the indicator shall be subjected to and meet the requirements specified in the Turn Indicator Sensitivity (Low Temperature) tests. The indicator shall be examined, and there shall be no damage which would adversely affect subsequent operation.

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4.5.21 *High Temperature Exposure.* (This test may be combined with the High Temperature Operation test at the discretion of the testing agency). The indicator, not operating and mounted in a level position, shall be subjected to a temperature of 70 ± 2°C (158 ± 3.6°F) for a period of 24 hours. At the end of this period, and while still at the high temperature, the indicator shall be subjected to and meet the requirements of the Turn Indicator Sensitivity (High Temperature) test. The indicator shall then be returned to room temperature and allowed to saturate for a minimum period of 4 hours. It shall then be subjected to and meet the requirements specified in the individual tests. The indicator shall be examined, and there shall be no damage which would adversely affect subsequent operation.

4.5.22 *Vibration Failure.* The indicator shall be mounted in a level position on a vibration stand and operated for 3 minutes. The indicator then shall be subjected to vibration as specified herein, with a circle between 0.018 and 0.020 of an inch. The frequency shall be varied at a uniform rate of 90 cycles per minute, (a variation of 2700 cycles), from 300 to 3,000 cycles per minute and from 3,000 to 300 cycles per minute, once each hour during this test. This cycle shall be repeated a sufficient number of times to extend the total time to approximately 3 hours (3 cycles). At the end of this period, the indicator shall be examined, and there shall be no looseness in the mechanism as a result of the vibration, and there shall be no damage which would adversely affect subsequent operation. The indicator shall be subjected to and meet the requirements specified in the Individual Tests.

4.5.23 *Humidity.* The indicator, with no power applied but external connections made to simulate installed conditions, shall be subjected to the Humidity tests; Procedure I, of Specification MIL–E–5272, except that it shall be for 120 hours (5 cycles). At the end of the 120 hour (5 cycle) period, within one hour after the indicator has been removed from the chamber, it shall be subjected to and meet the requirements specified in the Turn Indicator Individual Tests and the Dielectric Strength Test. Examination for corrosion and other deterioration shall be made between 24 and 48 hours after completion of the humidity exposure. There shall be no corrosion or other deterioration which would adversely affect subsequent operation of the indicator.

4.5.24 *Fungus Resistance.* The indicator, with no power applied, shall be subjected to the Fungus Resistance tests, Procedure I, of Specification MIL–E–5272. At the conclusion of this test, the indicator shall be examined, and there shall be no growth of fungus which will adversely affect subsequent operation of the indicator. The indicator shall be subjected to and meet the requirements specified in the Turn Indicator Individual tests.

4.5.25 *Salt Spray.* The indicator, with no power applied, but with external connections made to simulate installed conditions, shall be subjected to a salt spray in accordance with Specification MIL–E–5272 for a period of 50 hours. At the end of the 50 hour period, the indicator shall be examined and there shall be no corrosion or deterioration which would adversely affect subsequent operation of the indicator. The indicator shall be subjected to and shall meet the requirements specified in the Turn Indicator Individual tests and the Dielectric Strength Test.

4.5.26 *Mounting Lugs.* The indicator case shall be mounted face downward on the removable head of a suitable testing machine with the face of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine. A load of 175 pounds shall be applied for one minute to each lug in a direction toward the front of the case. The lugs shall withstand the applied load without fracture.

4.5.27 *Life.* The indicator shall be mounted

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in a normal position on a turntable. The indicator shall be subjected to a cycle consisting of starting and operating the indicator for a period of 2½ hours and then disconnecting the electrical power and allowing the instrument to cool off for a period of ½ hour. During the operating period of the indicator, the turntable shall rotate 20 degrees and return to its initial position 7 times per minute at a 360 degree per minute rate of turn. This cycle shall be repeated a sufficient number of times to extend the operating time of the indicator to a total of 1500 hours. There shall be no failure during the test and upon completion of this test the indicator shall be subjected to and meet the requirements specified in the Individual tests.

4.6 Rejection and Retests. When any representative sample fails to meet the requirements of the representative sample test, the lot represented shall be rejected and returned at the contractor's expense. Any instrument failing to meet the requirements of the Individual tests shall be rejected and returned at the contractor's expense. Indicators which have been rejected may be replaced or repaired to correct the defects and resubmitted for test. When this has been accomplished, all specified tests shall be repeated. Before resubmitting, full particulars concerning previous rejection and action taken to correct the original defects shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 Application. The requirements of Section 5 apply only to direct purchases by or direct shipments to the Government.

5.2 Level A. The preservation and packaging shall be in accordance with Specification MIL–P–6064, Group VIII, utilizing a Type II, Class I container.

5.2.1 *Level C.* The indicator shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.3 Packing.

5.3.1 *Level A and Level B.* Packing shall be in accordance with Specification MIL–P–6064.

5.3.2 *Level C.* Packages which require overpacking for acceptance by the carrier shall be packed in commercial exterior shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification Rules, or regulations of other common carriers, as applicable to the mode of transportation.

5.4 Marking of Shipments. Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL–STD–129. The identification shall consist of the following information listed in the order shown:

Stock No:

Indicator, Turn and Slip, 28V. DC, 2¾ Size, Type MD–10

MS–28069–1

6. NOTES

6.1 Intended Use. The electric turn and slip indicator covered by this specification is intended for use in aircraft to indicate the rates of rotation of the airplane about its vertical axis and the lateral attitude of the airplane relative to the apparent vertical.

6.2 Ordering Data. Requisitions, contracts and orders shall specify the type and part number of the indicator desired, the quantity, and whether overseas packing is to be furnished. (See Section 5)

6.3 Qualification. With respect to products requiring qualification, awards will be made

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only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date.

6.3.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief, Bureau of Aeronautics, Department of the Navy, Washington 25, D. C., the activity responsible for qualification.

Patent notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodians:

Army—Transportation Corps

Navy—Bureau of Aeronautics

Preparing activity:

Bureau of Aeronautics

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE PLATE NO. 15419

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